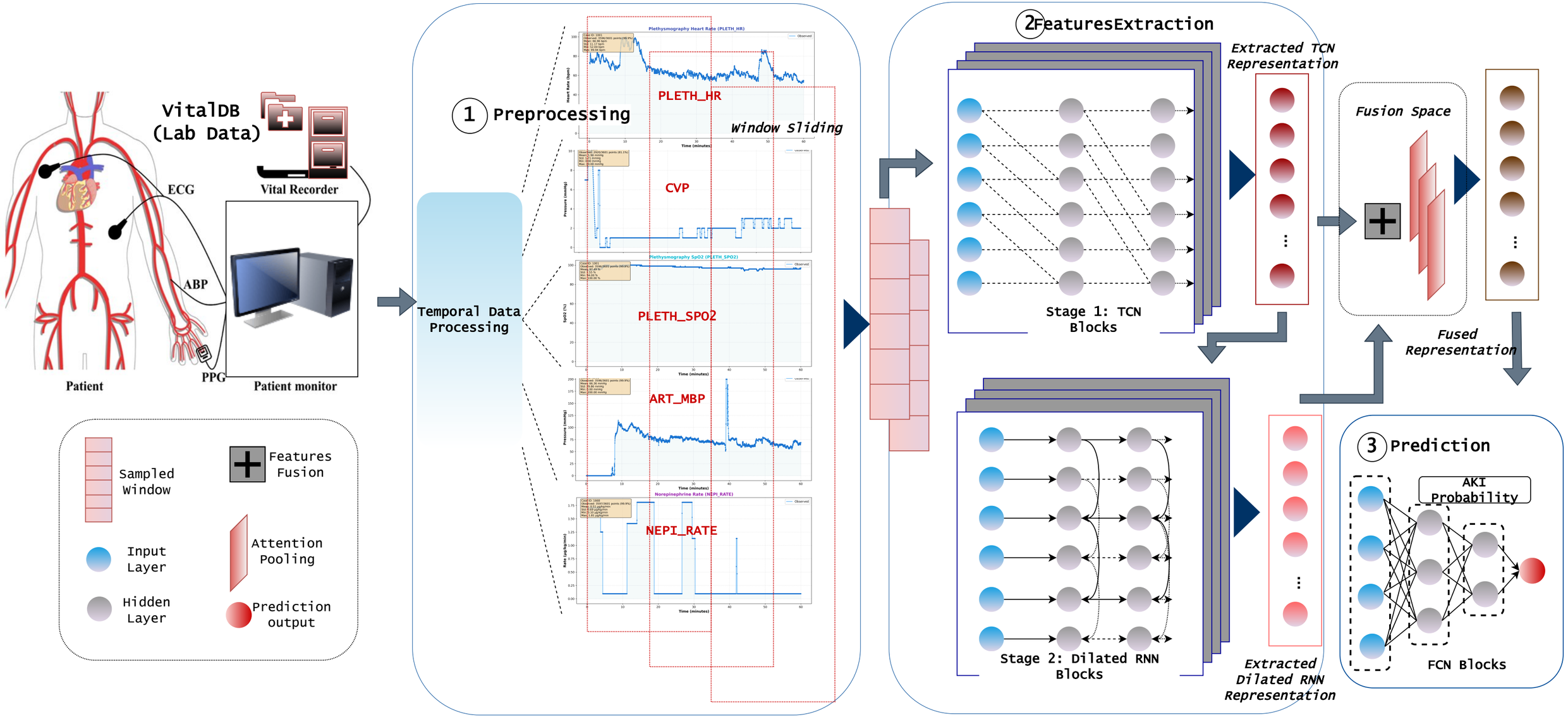
VitalDB
(Lab Data)
Vital Recorder
ECG
ABP
PPG
Patient
Patient monitor
Sampled Window
Features Fusion
Input Layer
Attention Pooling
Hidden Layer
Prediction output
1 Preprocessing
Temporal Data Processing
Window Sliding
PLETH_HR
CVP
PLETH_SPO2
ART_MBP
NEPI_RATE
2 FeaturesExtraction
Stage 1: TCN Blocks
Extracted TCN Representation
Fusion Space
Fused Representation
Stage 2: Dilated RNN Blocks
Extracted Dilated RNN Representation
3 Prediction
AKI Probability
FCN Blocks